



AbVance II

Focus on 'developability' risk prediction of antibody drug candidates

Partner proposal, October 2018

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Executive summary

Our confidence in the ability to predict antibody drug candidate ‘developability’ issues early by surrogate biophysical assays is limited by the low number of molecules with associated biophysical data and an agreed standard way of measuring this data across the industry. In addition, key pharmacological properties such as human and cyno PK (clearance), viscosity and solubility, and an understanding of whether these endpoint data could be predicted using biophysical surrogate data is lacking.

We are seeking endorsement for the creation of an industry-wide repository of ‘developability’ information on antibody drug candidates. This will provide greater efficiencies when accessing and querying data through consolidation, curation and standardisation of the information as well as a greater volume and diversity of information, including significant amounts of previously unavailable data on poorly behaved molecules and endpoint data. As a consequence, there will be greater potential for validation and refinement of existing predictive models of ‘developability’ risk and the generation of new models.

1 Introduction

Data is playing an ever greater role in the life sciences and across all drug discovery modalities including small molecules, antibodies & proteins. It is often said that data is now the true asset of drug discovery. More than ever the industry is looking towards employing big data and predictive analytics to improve R&D efficiency and help stay ahead in an increasingly competitive era which demands more patient focused drug development.

To pursue research projects effectively decisions need to be supported by access to increasingly vast stores of data. Some of the data will be derived from internally generated research efforts or external collaborations, and others from public and community-wide biomedical projects from around the world. The latter sources of data, posted in repositories across the Internet, are fast becoming a boon to experimentalists seeking to interpret their own results, as well as to the growing armies of computational teams & data scientists looking for patterns that only emerge by looking at the big picture gained from having access to larger cohorts of data. But, these massive datasets pose huge challenges per se; it can be difficult to find and download the datasets, interpret their formats and the details of the data, and perform computations combining these diverse information.

As such, there is not only a need for broad access to large numbers of biomedical data, but more than ever a need for ease of access, usability and overall confidence in the information on which key research decisions will be made. Thus, there is a need to bring together past and emerging data into core repositories under standard frameworks which will facilitate decisions within drug discovery projects going forward.

1.1 Current trends in prediction of assay data of antibody candidate drugs

The monoclonal antibodies (mAbs) market is expected to reach a value of approximately \$140bn by 2024; representing approximately 11% of the total worldwide prescription drug and OTC sales. The successful development of antibody molecules into approved drugs relies on optimizing a series of biophysical,

pharmacological and manufacturability properties before the start of clinical trials. Failure during clinical trials or during the associated later stages of development is expensive. As such, early prediction of 'developability' risk is of key importance in order to ensure the successful development of a lead candidate into a stable, manufacturable, safe and efficacious drug.

The use of surrogate screening assays and computational tools for the prediction of 'developability' parameters such as chemical and structural stability, aggregation, PK and viscosity is a potentially efficient and effective way to help rank candidate molecules. There has been large number of publications over the last 5 - 10 years, describing assays to predict different aspects of 'developability', many of which require relatively small amounts of protein (micrograms) to characterise and are designed to handle many candidate antibodies at once. These include, antibody solution properties, such as, solubility, viscosity and self-association, e.g. CIC, CSI-BLI, AC-SINS; physical/chemical stability, e.g. thermal stability by DSC/DSF, charge heterogeneity by cIEF; and assays to predict atypical clearance in vivo due to polyspecificity / nonspecific binding, e.g. baculovirus binding, PSR assay.

Despite this, our confidence in the ability to predict 'developability' risk early is limited by the low number of molecules with associated data and an agreed standard way of measuring this data across the industry. In addition, key pharmacological properties such as human and cyno PK (non-specific clearance), viscosity and solubility, and an understanding of whether these endpoint data could be predicted using biophysical surrogate data is lacking.

In the recent publication¹ by Jain and co-workers, twelve common biophysical assays (which include those mentioned above) were used to characterise 137 antibodies in advanced clinical stages. Analysis of the resulting data were used to quantify the historical limits of acceptability for certain 'developability' metrics and empirically define boundaries of drug-like behaviour for antibodies, akin to the Lipinski 'rule of five' for small molecules. Assays performed in the study were chosen based on there being a peer-reviewed publication of its use for therapeutic antibody characterisation, or it is a generally used assay (e.g., titers), and the assay being amenable to characterising hundreds of antibodies and consuming <1mg per assay of material. Based on these criteria there would be a likelihood of these or related assays being used in the early stages of antibody drug discovery.

Despite organisations using identical or similar assays to predict 'developability' risk and drive antibody discovery decision-making, it is common knowledge that each organisation follows internal protocols and emphasis is placed on certain assays over others even if these provide information on the same parameter, e.g. aggregation. Due to this underlying commonality among differently implemented assays of the same type there is an expectation that information derived from these assays can be directly compared between organisations.

1.2 Opportunities

Despite its undeniable impact, there are opportunities to build on the work conducted by Jain and co-workers to create further value:

¹ Jain et al. (2017) Biophysical features of the clinical mAb landscape. PNAS, 114 (5) 944-949; DOI:10.1073/pnas.1616408114

- (1) Expanding the dataset to include molecules from pre-clinical phases of drug discovery and including PK and other endpoint data would allow further analyses based on considerably more data points and correlations to be made between, e.g. PK and other parameters such as structural or chemical stability.
- (2) Share information on antibody molecules which display poor properties. All the molecules studied by Jain et al. had entered clinical trials and as such most display favourable properties. To gain a good understanding of 'developability' risk thresholds one could argue that a considerable number of molecules need to display sub-optimal properties.
- (3) Share information on 'real' molecules. The work done and conclusions derived by Jain et al are not based on the exact isotypes, let alone the full chain sequences, of the chosen antibodies. Despite characterisation of the actual drug product giving similar values in a small number of cases it is not possible to rule out that some of the data used in the analysis would not correspond to the that of the actual antibody molecule of interest.

2 Proposal

2.1 Vision, objectives and recommendations

2.1.1 Vision

Our vision is to create an industry-wide repository of biophysical and PK information for antibodies which will facilitate earlier selection of drug candidates.

2.1.2 Objectives

In achieving this vision we aim to meet the following objectives:

- (1) Establish the ability to compare relevant information (see section 2.3.1 for scope) across industry partners in a straightforward manner.
- (2) Identify a minimum set of 'value' assays which will provide sufficient information to run a 'standard' antibody drug discovery campaign.
- (3) Be able to better predict which molecules are likely to succeed in the clinic based on early biophysical data.
- (4) Predict ideal manufacturing and/or developability conditions (e.g. formulation) from early biophysical data.
- (5) Define which antibody sequence and 3D structure elements, and biophysical properties contribute to liabilities or undesired effects.
- (6) Be able to predict endpoint data (e.g. in vivo nonspecific clearance rates) from antibody sequence, 3D structure, Computational descriptors (see section 2.3.1 and Appendix II for more information) or a combination of these.

2.1.3 Recommendations

To meet the above objectives we recommend the following:

- (1) Consolidate information on antibody molecules into a single, curated data repository.
- (2) Share information on molecules with poor properties to help establish more accurate thresholds of 'developability' risk.
- (3) Identify common assays used by industrial partners and analyse the data from these to understand variability among assays, benchmark and establish best practices. Based on the work done by this group in preparation for this proposal, we have identified sixteen assays which are commonly used in industry

to drive progression from early screening to candidate drug selection (see Appendix I). Of these assays, only five are in common with those used in the work done by Jain et al.

(4) Further use the information in the repository to build in silico predictive models and as the basis for methods standardisation.

2.2 Value proposition and uniqueness

2.2.1 Value proposition

This initiative will result in the following for all participating members:

- (1) Greater efficiency when accessing and querying sequence, 3D structure, structure descriptors, biophysical and endpoint data through the consolidation, curation and standardisation of information on antibodies into a single repository.
- (2) Greater volume of information including endpoint (PK, Solubility, Viscosity) data available.
- (3) Greater diversity of information, including significant amounts of previously unavailable data and including data on poorly behaved molecules.
- (4) Greater potential for validation and refinement of existing in silico predictive models of ‘developability’ risk and the generation of new models.

2.2.2 Uniqueness

Organisations such as Lhasa² and Certara³ have developed software to model and predict aspects of metabolism and toxicology. Early last year, Certara announced the launch of a consortium which will use Quantitative Systems Pharmacology to predict the immunogenicity of biologics and its impact on pharmacokinetics, efficacy and safety in diverse patient populations. Lhasa hosts and manages all data shared by major pharmaceutical companies as part of the eTox⁴ project. The eTOX project was an Innovative Medicines Initiative (IMI)⁵, collaboration between 13 pharma companies, 11 academia institutions and 6 SMEs which shared historical toxicological data within the pharmaceutical industry. It created a series of models to support toxicity prediction. Both data and models are integrated in the platform developed in the project, the eTOXsys, which is a powerful system to access the eTOX data and the predictive models. ABIRISK is an IMI project focused on anti-drug immunogenicity. One of the first goals of the ABIRISK consortium was to provide clear definitions around terms and concepts related to immunogenicity, its prediction and associated clinical events.

To the best of our knowledge there are no initiatives aiming to compile sequence, structure, biophysical and endpoint information on antibodies from pharmaceutical companies with a view to better predict ‘developability’ risk. In relation to this proposal, Pistoia Alliance is facilitating a closely related initiative aiming to conduct a detailed literature review to extract well curated PK data on fully characterised antibody molecules in order to build a ‘gold standard’ set on which to base further work. This initiative resulted from the EMBL-EBI Industry programme workshop on PK prediction of Biologics held in June 2018. The initiative counts some of the authors of this proposal as members.

2.3 Approach

2.3.1 Scope

When addressing ‘developability’ risk prediction of antibody drug candidates, all information associated with antibody molecules, the types of data describing those molecules and the assays used to generate

² <https://www.lhasalimited.org/>

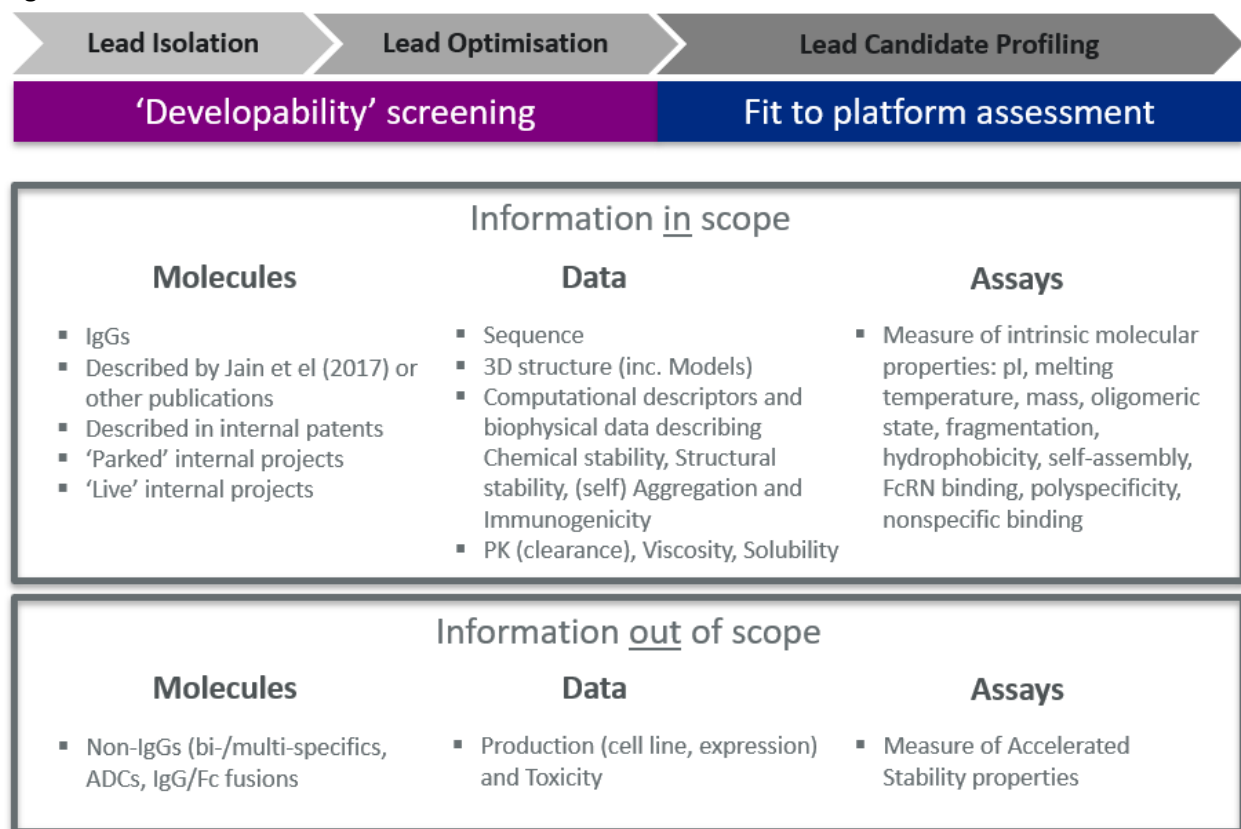
³ <https://www.certara.com/>

⁴ <http://www.etoxproject.eu/>

⁵ <https://www.imi.europa.eu/>

those data, from early screening to candidate nomination are effectively in scope. However, to test the value proposition (section 2.2.1) early we have chosen to limit the information sharing based on meeting our first objective (section 2.1.2), which is to understand whether we can compare similar discovery information on antibodies across partner organisations. To achieve this and to reduce the number of issues due to a perceived lack of comparability as well as variability issues we have chosen to focus on certain types of information. This is illustrated in summary form in Figure 1 and elaborated for 'Data' (to some extent) and 'Assays' in the Table found in Appendix I.

Figure 1



To avoid complexities between different molecule types we have chosen to focus on IgGs and their fragments e.g. Fabs. Due to perceived comparability reasons the inclusion of antibody production process information and assays which are used to generate Accelerated Stability data are out of scope (also see section 2.3.4.3).

The inclusion of sequence information is critical as this will allow the generation of 3D models, if crystal structures are not already available, and Computational and Biophysical-Molecular MOE⁶ descriptors (termed 'Computational descriptors' going forward - see Appendix II for examples). Computational descriptors on molecules of interest will be shared as an alternative to sharing sequence information, especially where this information is particularly sensitive. Computational descriptors will be used, amongst other things, to predict chemical and structural stability risk and help meet the objective to define which sequence and structure elements contribute to liabilities or undesired effects. We have

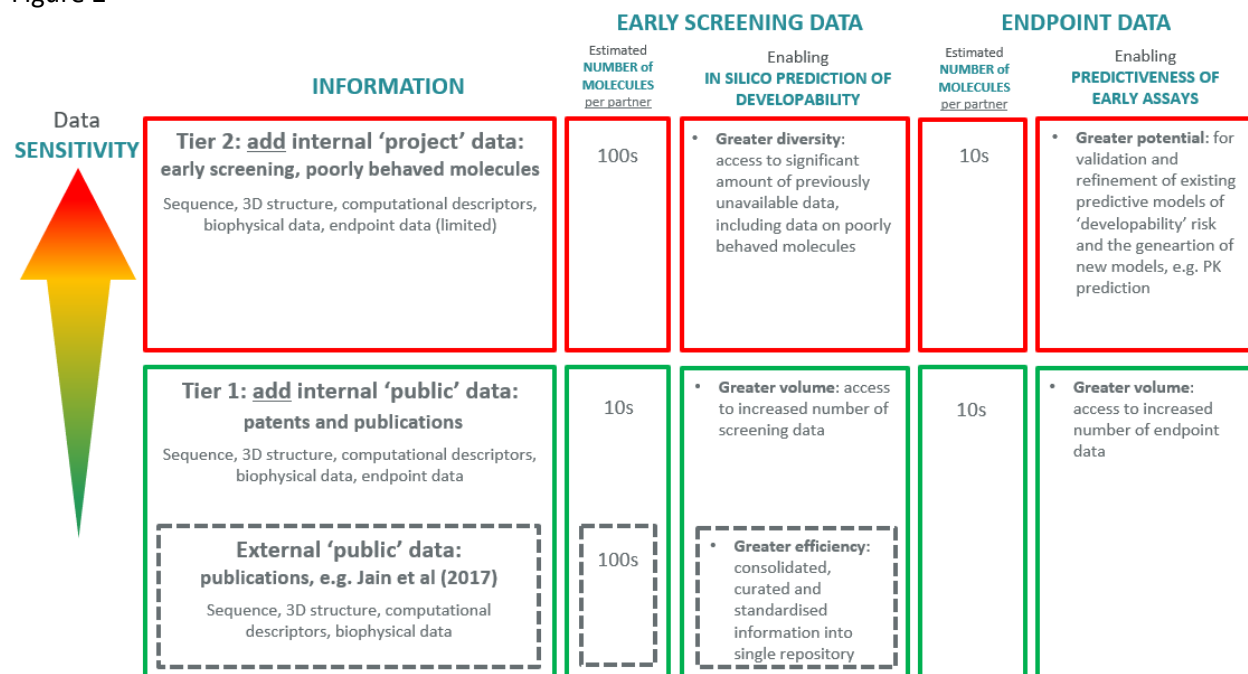
⁶ Molecular Operating Environment (MOE), 2013.8; Chemical Computing Group ULC, 1010 Sherbrooke St. West, Suite #910, Montreal, QC, Canada, H3A 2R7, 2018.

chosen not to include sharing of Toxicological data as this has been addressed by the eTox project and is also core business for Lhasa (see section 2.2.2).

2.3.2 Tiered approach and relationship to value proposition

In order to build value early and avoid IP issues associated with sharing proprietary, potentially sensitive, information we propose a tiered approach to sharing which leverages less sensitive information, that are readily accessible, at the start and subsequently incorporates information from more sensitive sources. The proposed tiered approach is depicted in Figure 2.

Figure 2



Tier 1 comprises all information in scope (section 2.3.1) derived from two main sources: (a) information which is currently available in the public domain and can be sourced from peer-reviewed publications such as Jain et al., (2017). We expect that information on 100s of molecules per partner are available from these sources and greater efficiencies in data querying alone would be achieved were this information to be consolidated into a single repository. Evolution of the repository through continued maintenance and expansion over time will guarantee continuing value. And, (b) information held within partner organisations related to molecules already in the public domain but which has not been released. We expect that additional information on some 10s of molecules per partner are available in this way. Previously unreleased endpoint information will add particular value but overall value is also achieved through greater volumes.

Tier 2 comprises all information in scope (section 2.3.1), held within partner organisations, on molecules which have not been published. Examples of sources for these types of molecules comprise; molecules from early screening campaigns which were not taken forward due to poor properties, molecules from projects in which the target is no longer of interest or molecules from ongoing projects. For the latter, if this information were to be shared, encryption and added security could be built as part of the repository to ensure the raw information is used in predictive model building but cannot be accessed and viewed by partners. Addition of Tier 2 information will enable access to unprecedented amounts of data from

different sources with our current estimates suggesting that this would be in the region of 100s of molecules per partner from early screening (Lead Isolation) and 10s of molecules per partner worth of Endpoint data.

As described above, and illustrated in Figure 2, our expectation is that the majority of novel data will fall under Tier 2 and come from molecules which were not progressed into Lead Optimisation/Lead Candidate profiling. When combined with Tier 1 data, this will provide a platform to develop predictive tools for early 'developability' risk prediction based on, e.g. (self) Aggregation data. The recent publication⁷ from Lonza Biologics presents the development and validation of such a (proprietary) tool based on data from over 500 antibodies. As mentioned earlier, the sharing of sequence and endpoint information is critical with regards the value proposition as ultimately one of our main objectives is to be able to predict non-specific clearance from sequence information. In working toward this objective, the availability of larger amounts of endpoint data will help improve the predictiveness of early screening assays for properties such as in vivo clearance, solubility, viscosity.

Ultimately, the creation of such a repository, which is in some ways akin to what the Protein Data Bank (PDB) is for protein crystal structures and associated data, will enable the development of computational and predictive models which are not yet envisioned. The recent publication⁸ by Man-Un Ung and co-workers describes the use of a Machine Learning (ML)-based algorithm to refine the current classification of protein kinase structures in the PDB into active and inactive conformations. Further analysis of the small molecules recognised by each conformation highlights conformation-specific substructures which can be used for the design of inhibitors with optimised pharmacological effects. The development of the ML tool described would not have been possible without the availability of a well populated, curated and standardised repository of structural information such as the PDB.

2.3.3 Proposed operation model

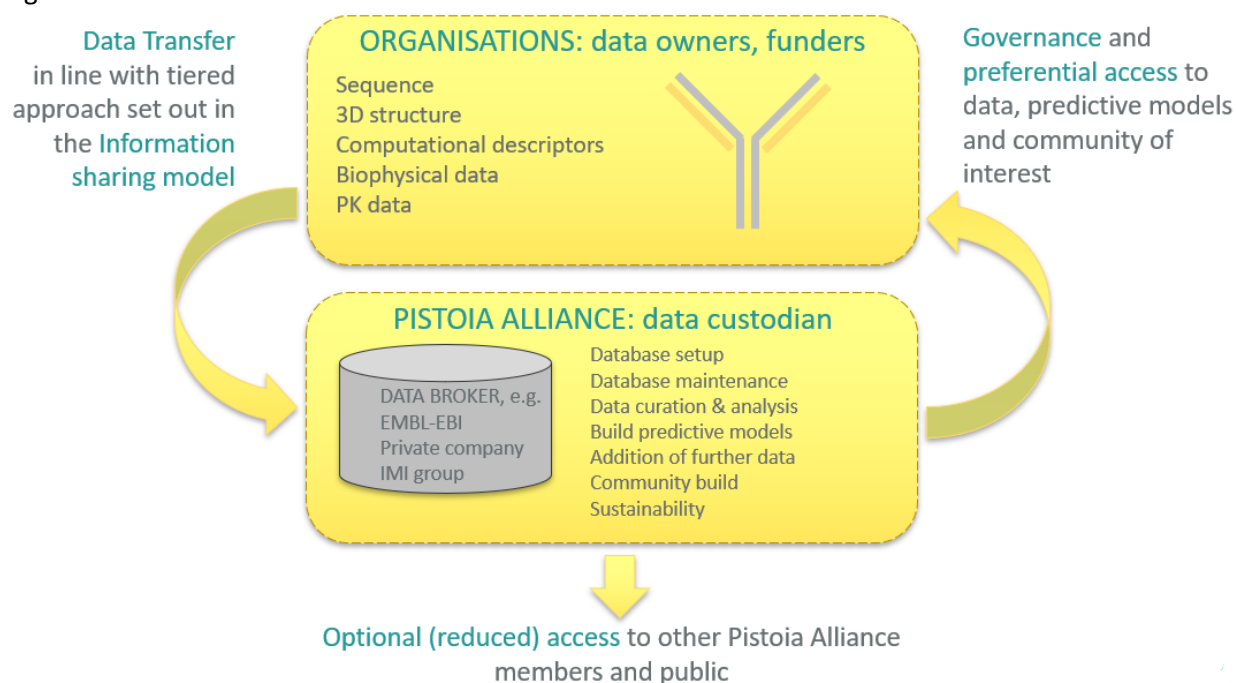
The proposed setup and operational model for enabling the creation and management of the data repository are illustrated in Figure 3. The table in Appendix III outlines the main roles and responsibilities in the proposed setup.

Partner organisations will contribute information into the database as described previously (sections 2.3.1 and 2.3.2) and will retain ownership of said information. Pistoia Alliance will act as the data custodian and project owner, thus being accountable for the management of the project which delivers the vision (section 2.1.1). Pistoia Alliance will appoint a Project Manager to lead the project. The project will comprise the setting up and maintenance of the data repository as well as the validation and generation of current and existing models of 'developability' risk respectively. In addition, a sustainability plan will be put in place to ensure future growth and value add of the data repository in line with the needs of partner organisations. All partner organisations will form part of the Steering Committee which will provide overall governance for the project and together with the Pistoia Alliance will be responsible for appointing a data broker to facilitate the deposition, curation and security of the information. All partner organisations will have exclusive access to the repository. Access to the repository by non-partners remains to be discussed.

⁷ Obrezanova et al. (2015) Aggregation risk prediction for antibodies and its application to biotherapeutic development. *MAbs*, 7 (2) 352-363; DOI:10.1080/19420862.2015.1007828

⁸ Man-Un Ung et al. (2018) Redefining the protein Kinase conformational space with Machine Learning. *Cell Chem Biol*, 25 (7) 916-924; DOI:10.1016/j.chembiol.2018.05.002

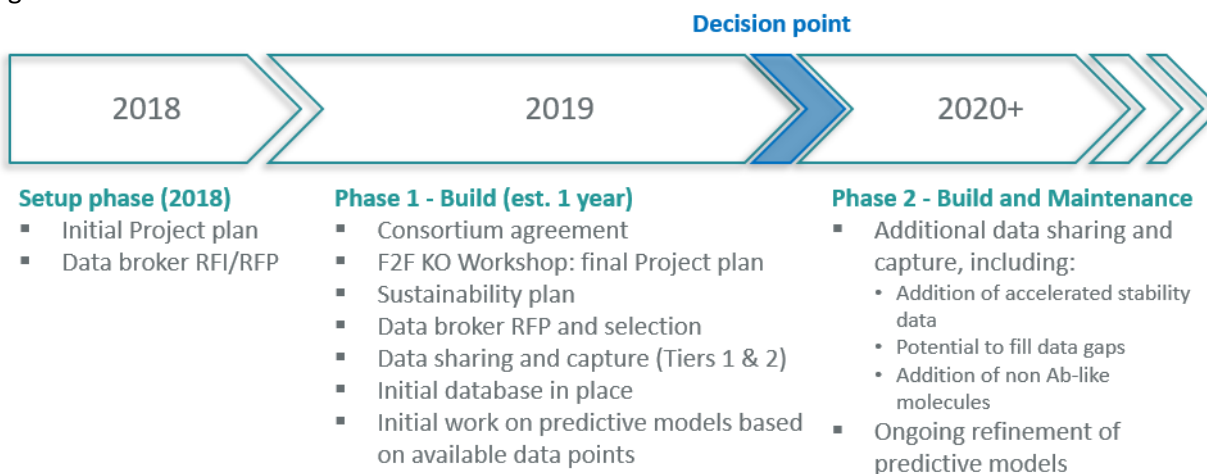
Figure 3



2.3.4 Plans

High level plans for the following 1-2 years are illustrated in Figure 4 and described in more detail in the following subsections.

Figure 4



2.3.4.1 Setup phase

This phase is currently ongoing and expected to last until the end of 2018. During this phase the main objective is to obtain approval for the initiative and build in specific requirements from internal stakeholders. Once approval has been sanctioned, Pistoia Alliance will begin the process of setting up governance and the legal framework to start the project with a view to completing this early in 2019. In addition, organisations which have the potential to act as data brokers will be engaged to assess their capacity to provide the required service. Finally, plans for the project face to face kick-off workshop to be

held in early 2019 will be put in place. Costs associated with the setup phase will be covered by the Pistoia Alliance AbVance initiative. For an overview on estimated costs see section 2.3.5.

2.3.4.2 Phase 1 (Build)

Phase 1 is expected to start early in 2019, last for one year and deliver a data repository containing information in scope (section 2.3.1, Figure 1). In addition, Phase I will deliver a sustainability plan which will include plans for Phase 2 (Build and Maintenance). Based on information provided by Lhasa, Phase 1 is expected to have higher costs associated with it than Phase 2 (depending on activities and duration). These higher costs are mainly associated with data entry but can be reduced based on data owners performing their own entry of data using standard schemas provided. At the end of Phase 1 a decision will be made whether to progress to Phase 2 with concomitant funding allocation to support this.

2.3.4.3 Phase 2 (Build and Maintenance)

Phase 2 is expected to follow seamlessly from Phase 1, starting in early 2020. The duration of Phase 2 will be largely dependent on a number of factors such as; achievements during Phase 1, sustainability strategy and allocated funds. There is an expectation that information currently deemed out of scope (see section 2.3.1, Figure 1) will be assessed for inclusion during Phase 2 based on the strategy set out in the sustainability plan.

2.3.5 Costs

Funds sought from participating members will cover all Phase 1 activities and are illustrated in Table 1. Data broker costs are based on rough, top level, estimates provided by Lhasa and are not conditioned by variables such as: number of participating organisations, scaling based on size of each organisation and distribution of tasks e.g. entering data. As such, and since data broker costs will amount to the largest share of the total costs, the figures provided should be regarded as illustrative and not final.

Table 1

Source	Activity	Total per partner organisation (based on current # participants)
Pistoia Alliance	\$60k Project Management \$5k Travel \$6.5k Pistoia Alliance overhead (@10%)	\$18k
Data Broker	\$240k Data repository build	\$60k
TOTAL		\$78K

2.3.6 Key Implementation Risks

Five high level risks have been identified (Table 2), two of these would likely impact phase 2 but are included for awareness. A detailed risk assessment will be conducted as part of the project KO workshop.

Table 2

Risk No.	Description of Risk & likelihood	Impact (Phase) & Proposed mitigation measures
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R1	Delays to database creation due to Tier 2 (and some Tier 1) information sharing occurring on case by case basis. High	Medium/High (Phase 1). Request blanket approval based on set of required criteria. Make this a specific question to stakeholders.
R2	No sequence data. Sequence and/or structure information cannot be shared. High	High (Phase 1). Agree and use a set of Computational descriptors instead (see section 2.3.1 and Appendix II) or alternatively encrypt sequence information such that it can be incorporated into in silico model creation but is not visible to participants.
R3	Low comparability of data across partner organisations. Medium	High (Phase 1). The majority of assays identified are deemed to have 'high' comparability. If this were not feasible, 'medium' and 'low' categories can be used instead. Alternatively, control molecules (standards) chosen across institutions can be used to establish assay comparability or data can be generated externally at a single site (this approach would incur extra costs).
R4	Gaps in information for Tier 1 & Tier 2 molecules. Many of the assay workflows at certain organisations are relatively new with regards the molecules which have been patented. As such, assay data is likely to be missing for some of the highlighted assays. High	Medium (Phase 2). Generate missing data internally or through CRO using new workflows at extra cost.
R5	Reluctance to generate new data on clinical candidates (Tier 2) or marketed molecules (Tier 1) as new data may highlight unfavourable properties. Medium	Low (Phase 2). Limit generation of new data to molecules which are not in the clinic or on the market - this includes largest number of molecules.

3 Summary

Through this proposal we are seeking endorsement for the creation of a unique, industry-wide repository of 'developability' information on antibody drug candidates. The repository will provide a greater volume and diversity of data, drive efficiencies in accessing relevant data and offer a greater potential for building more accurate models for predicting 'developability' risk.

To achieve this value proposition; sequence, 3D structure, Computational descriptors, biophysical data describing chemical and structural stability, aggregation and Immunogenicity data, as well as endpoint data (PK, viscosity and solubility) needs to be shared by all partners involved. Our preliminary work suggests that comparability, for the majority of common biophysical and endpoint assays identified, would be high. Sharing of unpublished information, especially for early screening data, will begin to provide the volume of information needed to build further accuracy and novelty to in silico predictive models of 'developability'. In addition, the sharing of sequence and endpoint information is critical as these data would enable realistic efforts to predict endpoint data from antibody sequence, 3D structure, and computational descriptors.

Once established, the expectation is that the repository will grow and evolve into an invaluable resource for participating organisations and that through the governance provided by its participants will provide value beyond that currently envisioned in this proposal.

Appendix I - Assay Table

Heading keys:

Data needed

Minimum information to be shared for each assay or molecule in addition to that described in section 2.3.1, Figure 1.

Impact of production process;

'Likely' <define cause of potential impact or reason>, or 'Unlikely'

Likelihood of comparability;

Includes assay clusters which observe the same phenomenon's, e.g. DSF and DSC

definitions (assumption that individual assays within the same organisation are reproducible);

'High' = deemed possible to directly compare absolute values across assay clusters which observe the same phenomenon's e.g. DSF & DSC.

'Medium' = deemed possible to directly compare absolute or relative (with respect to appropriate controls) values between individual assays.

'Low' = deemed possible to compare individual assays only using Red, Amber, Green flags (categories to be defined).

Parameter measured	Assay name	Assay abbreviation	Assay description (link to primary citation)	Purpose	Tech platform, e.g. NanoPro	Used by <company>	Stage	Measure of (units)	Data needed	Impact of production process	Likelihood of comparability	Additional information or comments
INTRINSIC MOLECULAR PROPERTIES												
Chemical stability	Capillary Isoelectric Focusing	cIEF	High throughput, capillary-based isoelectric immunoassay to determine charge profile and pI of molecule (Michels et al., Anal. Chem 2012 dx.doi.org/10.1021/ac3008847)	Measure of pI and charge heterogeneity - used in conjunction with LC-MS.	NanoPro Beckman Proteinsimple	GSK Lilly MedImmune	Lead Isolation Lead Optimisation Lead Candidate profiling	pI	Buffer conditions pI of main isoform Proportion of main isoform to acidic and basic species	Low for antibodies (May be significant for other proteins in view of glycosylation impact)	High - Should be comparable even with subtle differences in method and antibody formulation	Parameter of low value - can be predicted in silico reliably enough from sequence (published)
Chemical stability, Immunogenicity	Liquid chromatography-Mass spectrometry	LC-MS	Analytical technique used to separate isoforms by liquid chromatography (LC) and determine their mass by spectrometry (MS)	Intact molecular mass and 'standard' glycosylation pattern and obvious fragmentation.		Abvance GSK Lilly MedImmune Novartis	Lead Optimisation Lead Candidate profiling	MW % intact species % of respective glycovariants	MW of main isoform Buffer conditions	Likely - Glycan composition will affect 'standard' pattern	High	Parameter of low value - intact molecular mass and fragmentation can be assessed by SDS-PAGE. Ranking of molecules by LC-MS and CE-SDS comparable. Immunogenicity, need to define what a 'standard' glycosylation

												pattern is and how this is linked to immunogenicity
Structural stability	Differential Scanning Fluorimetry (including ThermoFluor and Nano DSF)	DSF	DSF is a fast and economical method to determine the temperature at which a whole antibody or a Fab unfolds (T _m), i.e. the temperature at the mid-point of the unfolding transition. The T _m is measured by an increase in the fluorescence of a dye with affinity for hydrophobic parts of the protein, which become progressively more exposed as the antibody unfolds. The T _m is assigned using the first derivative of the raw fluorescence emission data.	Rank Fabs and antibodies according to their thermal stability as measured by the T _m . Screen for favourable buffer and/or additive conditions.	iQ5, CFX96 Real-Time Systems (Bio-Rad) Prometheus (Nanotemper)	Abvance GSK Lilly (rarely) Medimmune Novartis (used by Jain et al)	Lead Isolation	Thermal stability (°C)	Buffer conditions Number of unfolding transitions Dye used, as this can affect T _m (e.g. SPYRO Orange) Conc of sample, as will affect T _m Temp gradient used during run (e.g. A temperature ramp from 40 °C to 95 °C at a rate of 0.5 °C/2 min)	Unlikely	High - assuming similar buffer conditions	Ranking of molecules by DSF and DSC comparable
Structural stability	Differential Scanning Calorimetry	DSC	DSC is a thermoanalytical technique in which the difference in the amount of heat required to increase the temperature of a sample vs. reference is measured as a function of temperature.	Rank Fabs and antibodies according to their thermal stability as measured by the T _m . Screen for favourable buffer and/or additive conditions.		Lilly Medimmune Novartis	Lead Optimisation	Thermal stability (°C)	Buffer conditions Number of unfolding transitions Dye used, as this can affect T _m (e.g. SPYRO Orange) Conc of sample, as will affect T _m Temp gradient used during run (e.g. A temperature ramp from 40 °C to 95 °C at a rate of 0.5 °C/2 min)	Unlikely	High - assuming similar buffer conditions	Ranking of molecules by DSF and DSC comparable
Structural stability, (self) Aggregation	Analytical Size-Exclusion Chromatography	aSEC	Antibodies are flowed through chromatography media consisting of non-aggregated dimeric and monomeric antibodies elute as defined, well-separated peaks. Aggregated antibodies flow through the column faster and elute before the non-aggregated antibody peaks. Percentage aggregation can be worked out from the ratio of peak surface areas under the aggregated vs non-aggregated antibody peaks.	Verify structural integrity, oligomeric state, percentage aggregation of antibody samples.	GE Healthcare Agilent	Abvance GSK Lilly Medimmune Novartis (used by Jain et al)	Lead Isolation Lead Optimisation	Elution time Peak width Peak asymmetry relative to a well-behaved control antibody (% monomeric species, retention time per unit volume or time)	Buffer conditions Capture column type (e.g beads 25-45 µm in size - GE Superdex 200). Flow rate Relative size (MW)	Unlikely	High	

Chemical stability (integrity), Structural stability (integrity)	Capillary electrophoresis	CE-SDS	Fast and high resolution analytical technique used to separate molecules within an electric field based on charge, viscosity and hydrodynamic radius.	Verify chemical and structural integrity, quantify fragmentation	Beckman Caliper	GSK Lilly MedImmune Novartis	Lead Optimisation Lead Candidate profiling	Approx. MW Impurities	Sample preparation conditions: reduced, non-reduced, denaturation	Unlikely	High	Ranking of molecules by LC-MS and CE-SDS comparable
(self) Aggregation (Hydrophobicity)	Hydrophobic Interaction Chromatography	HIC	HIC separates antibodies on non-biological surfaces based on the distribution and nature of the antibody hydrophobic patches. This approach assumes that the increased retention of antibodies on hydrophobic columns at moderate to high salt concentrations (e.g. ammonium sulphate, ammonium phosphate) correlates with increased antibody hydrophobicity. Antibodies are eluted in a mobile phase without added salt, i.e. only in buffer.	Assess non-specific interactions. Rank antibodies according to their relative hydrophobicity	GE Healthcare Agilent	GSK Lilly MedImmune Abvance Novartis (used by Jain et al)	Lead Isolation	Retention time (retention time per unit volume or time)	Buffer conditions Column type (e.g beads 25-45 μ m in size - GE Superdex 200). Salt gradient	Unlikely	High - should be comparable (relative retention time rather than absolute values)	Retention time used for ranking of molecules.
Structural stability, (self) Aggregation	Dynamic Light Scattering	DLS	Dynamic Light Scattering (DLS) measures the translational diffusion coefficients D_t of nanoparticles and colloids in solution by quantifying dynamic fluctuations in scattered light due to Brownian motion of the molecules in liquid. Sizes and size distributions, in turn, are calculated from the diffusion coefficients in terms of hydrodynamic radius r_h or hydrodynamic diameter d_h . Technique can be used with high concentration samples (<i>cf</i> SEC)	Measure hydrodynamic radius (characterize and quantify aggregation)	Wyatt	Lilly (rarely) MedImmune	Lead Optimisation Lead Candidate profiling	Hydrodynamic radius (nm)	Buffer conditions Sample concentration	Unlikely	High - should be comparable for same buffer conditions and sample concentrations	Ranking of molecules by DLS and AC-SINS comparable
(self) Aggregation	Self-interaction by light scattering	K-Diff	DLS experiment which measures the propensity for molecule to self-associate under particular conditions. K-diff = Association constant defined by diffusion.	Rank antibodies according to their propensity to self-assemble	DLS	Lilly MedImmune Novartis	Lead Optimisation	(mL/g)	Buffer conditions Concentration range K-diff	Unlikely	Probably High	

(self) Aggregation	Affinity-Capture Self-Interaction Nanoparticle Spectroscopy	AC-SINS	High-throughput, low protein screen for self-association propensity, where antibodies are captured onto gold nanoparticles. Self-association is determined if the nanoparticles with captured antibody clump together resulting in a red-shift. Sule et al., Mol. Pharmacuetics, 2013 dx.doi.org/10.1021/mp300524x	Assess propensity of antibody to self-associate / RSA at high concentrations	Gold nanoparticles Scanning plate reader	Medimmune (used by Jain et al)	Lead Isolation	Plasmon wavelength redshift in nm	Buffer conditions Record redshift in nm, relative to non-capture / buffer only controls	Unlikely	High - should be comparable for same buffer conditions	
(self) Aggregation	Multiangle laser light scattering	MALLS	MALLS describes a technique for measuring the light scattered by a sample into a plurality of angles. It is used for determining both the absolute molar mass and the average size of molecules in solution, by detecting how they scatter light. (wikipedia)	Determine symmetry or presence of multiple aSEC peaks and shape independent measure of MW		Abvance						Does not provide much additional information over aSEC experiment
Chemical stability, Structural stability (Immunogenicity)	In silico predictions	n/a	Application of computational tools focused on predicting chemical degradation sites, aggregation and immunogenicity of antibodies	Predict pI and PTM sites in CDR		GSK Medimmune Novartis	Lead Isolation		Primary sequence Crystal structure (if available) or model Computational descriptors	n/a	High	Not of high interest for comparisons. Can be applied to all selected molecules easily based on available sequence.
human, cyno, mouse (GSK), rat (Novartis) PK		FcRN interaction (SPR)	Binding to FcRn contributes to the long half-life of IgGs and albumine. Properties of antibodies to bind FcRn receptors is assessed for candidate ranking and risk minimizing.	Assessment of FcRn binding properties to identify risk of unfavorable PK.	Biacore EPIC	GSK Lilly Novartis	Lead Optimisation Lead Candidate profiling	Apparent affinity at pH 6.0, 7.4 to FcRn (M) % Residual binding to FcRn observed	Buffer conditions Assay setup (ratio Ab:receptor, fitting model) and readout	Unlikely	Medium/Low - vs control, not absolute numbers as dependent on conditions used	Not necessarily predictive - unusual activity for FcRN signifies potential issue. However, in some cases no FcRN activity results in abnormal PK.
(human, cyno) PK	Heparin Sulphate Chromatography		Mammalian cell membranes are formed from negatively charged components such as heparin. Non-specific binding to heparin-coated plates provides an indication of fast non-specific clearance. (Datta-Mannan et al., Balancing charge in the complementarity determining regions of humanized mAbs without affecting pI	Rule out proteoglycan interactions affecting PK	HPLC (Agilent, Waters, Thermo, etc)	Lilly	Lead Isolation Lead Optimisation	Retention time (or salt concentration at elution), relative to control antibodies	Buffer conditions Column type (e.g beads 25-45 µm in size - GE Superdex 200). Salt gradient	Unlikely	High	Not standard use. Data comparable to heparin binding (in cells)

			reduces non-specific binding and improves the pharmacokinetics. MAb 2015; 7(3):483-93; PMID:25695748; https://doi.org/10.1080/19420862.2015.1016696									
(human, cyno) PK	ELISA Baculovirus particles	BVP ELISA	ELISA on baculovirus particles and non-coated ELISA plates under low stringency blocking conditions to assess nonspecific binding / polyspecificity of lead antibody candidates. Binding observed to either BVPs or plastic indicates that the antibody has a risk of faster nonspecific clearance in vivo. Hotzel et al., mAbs, 2012 https://www.tandfonline.com/doi/abs/10.4161/mabs.22189	Polyspecificity / nonspecific binding / aberrant PK risk	ELISA	MedImmune (used by Jain et al)	Lead Isolation Lead Optimisation	Absorbance at 450nm if using HRP detection	Buffer conditions Concentration range of antibodies tested Absorbance on both BVPs and non-coated wells Record binding ratio relative to controls (sticky and non-sticky)	Unlikely (providing there is high monomer purity)	Low - absolute numbers not comparable but whether an antibody flags should be comparable.	Low use across industry - possibly used by Adimab and Genentech
(human, cyno) PK	Nonspecific HEK or CHO cell binding by flow cytometry (HyperCyte) or non-wash cytometry (Mirrorball)	HEK/CHO binding	Providing the target for an antibody panel is not expressed on a certain cell type (e.g. HEK293 or CHO), then assay systems using this cells are effective in determining the polyspecificity and therefore PK risk.	Polyspecificity / nonspecific binding / aberrant PK risk	Flow cytometry, Mirrorball (TPP Labtech)	MedImmune	Lead Isolation Lead Optimisation	Fluorescence counts	Assay conditions Concentration range of antibodies tested Total fluorescence counts Fluorescence ratio relative to controls (sticky and non-sticky)	Unlikely (providing there is high monomer purity)	Low - absolute numbers not comparable but whether an antibody flags should be comparable.	
Solubility	Solubility	Ultra-filtration	Ultrafiltration (centrifugal) is used to concentrate a protein to its maximal solubility limit; the protein concentration is then determined by absorbance.	To determine the solubility (and physical stability) of an antibody in specific conditions % of original concentration retained under specific conditions	Ultrafiltration	GSK Lilly Novartis	Lead Candidate profiling. Lead Optimisation	mg/mL	Absolute concentration % concentration vs original (absolute) Buffer conditions	Low likelihood (only in the event that glycan distribution affects solubility properties)	High (when measured in similar buffer conditions)	
Viscosity	Viscosity		Measure of viscosity (resistance to deformation) of antibody solution based on flow alone (viscosity) or flow based on applied force (rheometry)	Assess viscosity of unformulated vs formulated antibody	Malvern Rheometer Malvern Viscosizer	GSK (not routine so limited data) Lilly MedImmune Novartis	Lead Candidate profiling		Buffer conditions Concentration	Unlikely	Low (High - if measured in similar buffer conditions)	

Appendix II - Examples of Computational and Biophysical-Molecular MOE descriptors

Tables are not extensive or entries chosen due to importance, but for illustrative purposes. A prioritised list will be put in place ahead of the project start.

Computational descriptors

Publication date	Title	Author affiliation	Descriptors/algorithm	Dataset
J Pharm Sci. 2012 Jan	Developability index: a rapid in silico tool for the screening of antibody aggregation propensity (DI)	Novartis & MIT	DI; SAP, net charge	12 mAbs
PNAS 2014 Nov 17	In silico selection of therapeutic antibodies for development: Viscosity, clearance, and chemical stability	Genentech	Fv charge, HI, FvCSP	61 mAbs
Pharm Res. 2014 May	Concentration dependent viscosity of monoclonal antibody solutions: explaining experimental behavior in terms of molecular properties	Pfizer	net charge, pl;	11 mAbs
Bioinformatics 2017 Dec	Prediction of delayed retention of antibodies in hydrophobic interaction chromatography from sequence using machine learning	Adimab	ML; Adimab SASA	10,000 exp. data

Biophysical-Molecular MOE descriptors

Code	Class	Description
Atom Clashes	Protein	Atom clash count (> 0.5kcal/mole) as from Homology Model
BB Angle Outliers	Protein	Backbone bond angle outliers as from Homology Model
BB Bond Outliers	Protein	Backbone bond length outliers as from Homology Model
BB Torsion Outliers	Protein	Backbone bond torsion outliers as from Homology Model
Contact Energy	Protein	Effective contact energies as from Homology Model
E bond	Protein	The bonded energy (including environment atoms)
E ele	Protein	The electrostatic energy (including environment atoms)
E sol	Protein	The solvation energy (including environment atoms)
E vdw	Protein	The van der Waals energy (including environment atoms)

Appendix III - Roles, responsibilities and governance model

Role	Description	Who
Project owner	Accountable for the management of the project (AbVance II) which delivers the vision.	Pistoia Alliance. Pistoia Alliance will appoint a Project Manager to lead the project.
Data owner	Retains IP over the data in question and commits to releasing the data into the database as part of the project.	Organisation which generated the data in question.
Data custodian	Retains IP over the database and any outputs e.g. predictive models. Accountable for managing the deposition, curation, safety of the data and its use e.g. for the generation of predictive models.	Pistoia Alliance. Pistoia Alliance will appoint a Data broker (see below) in consultation with the Steering Committee (see below).
Data broker	Responsible for the deposition, curation, safety of the data.	Organisation with the right expertise and capabilities to set up, incorporate all data into, and maintain the data repository.
Consortium member	Members of the Pistoia Alliance which have an interest and expertise in developability risk prediction.	Pistoia Alliance, data owners and any organisation which has contributed funds as part of the project.
Steering Committee member	The Steering Committee provides overall governance for the project which delivers the vision.	Consortium members will appoint a relevant person to the Steering Committee. The Steering Committee will be chaired by Pistoia Alliance through the Project Manager.
Project team member	Project Team members provide required expertise to deliver aspects of the project and together will be responsible for the overall delivery of the project.	Consortium members and the data broker will appoint a relevant person to the Project Team. The Project Team will be led by the Project Manager.